Yassir Boulaamane

Postdoctoral researcher
InSiliChem, Universitat Autònoma de Barcelona

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Website · LinkedIn · GitHub · Google Scholar · ORCID
Moroccan · Spanish work permit



Profile

PhD in cheminformatics, computational chemistry, molecular modelling, and machine learning. The work covers natural-product drug discovery for neurodegenerative and infectious diseases, and cross-species cytochrome P450 metabolism prediction. Currently seeking roles in computational drug discovery.

Experience

Apr 2026
Postdoctoral researcher, secondment
Syngenta Group · London, United Kingdom
- Designed deep learning models for cytochrome P450 metabolism prediction.
- Worked with an international scientific group on data-driven product-safety pipelines.

Sep 2025 – present
Postdoctoral researcher
Universitat Autònoma de Barcelona · Barcelona, Spain
- Molecular modelling of metabolizing cytochromes P450 for a new generation of pesticides.
- Predictive models of pesticide reactivity with cytochromes P450.
- Deployment of cross-species metabolism models for safety, selectivity, and sustainability.

Nov 2024 – Dec 2024
Associate researcher
Future Therapeutics · Berlin, Germany, remote
- Computational drug-discovery projects as an independent associate.

Oct 2024 – Nov 2024
Adjunct instructor
Abulcasis International University of Health Sciences · Rabat, Morocco
- Scientific methodology teaching and supervision of medical thesis writing.

Sep 2024 – Jul 2025
Research associate
Abdelmalek Essaadi University · Tangier, Morocco
- Undergraduate lectures and seminars, and biochemistry laboratory sessions.
- Supervision of bachelor's and master's theses.
- Drug-discovery projects on neurodegenerative disease and antimicrobial resistance.

May 2024 – Jun 2025
Pharma consultant
Pharma IT · Copenhagen, Denmark
- Configured and validated Veeva Vault Clinical and Quality applications.
- Dry-run and regression testing for compliance and functionality.
- SOP writing for regulated processes, and support for CTD drafting.

May 2023 – Jul 2023
Visiting PhD student
Andalusian Center for Developmental Biology, Universidad Pablo de Olavide · Seville, Spain
- Antibacterial assays: microdilution, time-kill, permeability, and checkerboard.
- Outer-membrane protein profiling and bacterial adhesion studies, with Prof. Younes Smani.

May 2022 – Oct 2022
Visiting PhD student
Department of Environmental Engineering, University of Patras · Agrinio, Greece
- DNA extraction and PCR from insect samples, and metagenomic microbial profiling.
- NGS library preparation and analysis with Prof. George Tsiamis.

Education

Dec 2020 – Jul 2024

PhD, computational drug discovery

Abdelmalek Essaadi University · Tangier, Morocco

Highest honors, EQF 8. Thesis: Data-driven discovery of bioactive natural products: application to neurodegenerative and infectious diseases. Supervised by Prof. Amal Maurady.

Sep 2018 – Jul 2020

MSc, Biology and Biomaterials

Mohammed V University in Rabat · Rabat, Morocco

Good, EQF 7. Thesis: Insights into the structure–activity relationship of alkynyl-coumarinyl ethers as selective MAO-B inhibitors using molecular docking.

Sep 2012 – Jul 2017

BSc, Biology and Health

Mohammed V University in Rabat · Rabat, Morocco

EQF 6. Thesis: Health impacts of glyphosate exposure.

Skills

Modelling and data

Molecular docking and virtual screening · QSAR models · Machine learning · Deep neural networks, including CNNs and GANs · Graph neural networks · GROMACS molecular dynamics · HyperChem, UCSF Chimera, Avogadro, AutoDock Vina

Computing

Python · Bioinformatics tools · HPC environments and the SLURM scheduler

Laboratory

Basic microbiology methods

Languages

French. **C1** across listening, reading, spoken production, spoken interaction, and writing
English. listening **B2**, reading **C1**, spoken production **B2**, spoken interaction **B2**, writing **C1**
Arabic. Mother tongue

Talks and posters

Organizing committee, International Bioinformatics Conference, Tangier, 2021–2024.

Sep 2026

Poster

EFMC Medicinal Chemistry 2026 · Basel, Switzerland

Combined enhanced conformational sampling and protein-ligand dockings on CYP3A4 improves in silico prediction of drug metabolism

Dec 2024

Talk

International Bioinformatics Conference IBC'2024, Hotel Solazur · Tangier, Morocco

Computational screening of natural products as tryptophan 2,3-dioxygenase inhibitors: Insights from machine learning QSAR, molecular docking, ADMET, and molecular dynamics simulations

Oct 2024

Poster · Best ePoster

International eHealth Forum 2024, Mohammed VI Foundation for Sciences and Health · Casablanca, Morocco

Antibiotic discovery with artificial intelligence for the treatment of Acinetobacter baumannii infections

Oct 2023

Talk

International Bioinformatics Conference IBC'2023, Hotel Solazur · Tangier, Morocco

Enhanced accuracy in predicting drug blood-brain barrier permeability with a Machine Learning Ensemble model

Oct 2022

Talk

International Bioinformatics Conference IBC'2022, Hotel Solazur · Tangier, Morocco

QSAR and molecular modeling studies for the discovery of natural products as multi-target-directed drugs for Parkinson's disease

Dec 2021

Poster · Best poster

International Bioinformatics Conference IBC'2021, Hotel Solazur · Tangier, Morocco

Machine Learning model to predict potential Monoamine Oxidase B inhibitors from Cannabis Compound Database

Dec 2021

Talk

International Bioinformatics Conference IBC'2021, Hotel Solazur · Tangier, Morocco

Computational studies of African Natural Products Databases to identify natural dual-target-directed antiparkinsonian drugs

Jun 2021

Talk

1st International Bioinformatics Conference (iABC2021) · Istanbul, Turkey

Docking-based virtual screening and ADME evaluation of caffeine-based phytochemicals as inhibitors of Monoamine Oxidase B

Awards

Oct 2024

Best ePoster

International eHealth Forum 2024

Dec 2021

Best poster

International Bioinformatics Conference 2021

Publications

2026 Molecular Dynamics-Guided Triage of Approved Drugs Against Escherichia coli Resistance Targets Nominates Capmatinib, Metergoline and Nilotinib as Candidates for Experimental Testing
Mena Abdelsayed et al. · International Journal of Molecular Sciences · DOI

2026 Combined Enhanced Conformational Sampling and Protein-Ligand Dockings on CYP3A4 Improves in Silico Prediction of Drug Metabolism
Y. Boulaamane et al. · ChemRxiv · preprint, first author · DOI

2026 Integrative computational repositioning of Foretinib and PF-03715455 as potential GSTP1 inhibitors for cancer chemosensitization
Marouane Aherkou et al. · Scientific Reports · DOI

2026 QSAR-Guided Virtual Screening and Molecular Dynamics Reveal Olaparib as a Repurposing Lead Against α-Synuclein Aggregation
Mena Abdelsayed et al. · International Journal of Molecular Sciences · DOI

2026 FDA-Approved Drug Repurposing as p53 Mutants Rescue Candidates Using Structure-Based Virtual Screening and Molecular Simulations
Mena Abdelsayed et al. · International Journal of Molecular Sciences · DOI

2026 Repurposing FDA-Approved Drugs as Nav1.7 Channel Modulators: An Integrated Structure-Based Virtual Screening and Molecular Dynamics Study
Mena Abdelsayed et al. · International Journal of Molecular Sciences · DOI

2026 ARumenamides as Multitarget Ion Channel Modulators: Insights from Fenestration-Focused Docking, ADMET Profiling, and Molecular Dynamics
Mena Abdelsayed et al. · International Journal of Molecular Sciences · DOI

2026 Odorant-Binding Protein Interactions with Herbivore-Induced Volatiles Drive Behavioral Attraction of Harmonia axyridis (Coleoptera: Coccinellidae) to Tuta Absoluta-Infested Tomato Plant
Tijjani Mustapha et al. · Journal of Agricultural and Food Chemistry · DOI

2026 Repurposing DrugBank compounds as NAD-dependent deacetylase sirtuin 2 inhibitors via QSAR modelling with gradient boosting algorithms and all-atom molecular simulations
Y. Boulaamane et al. · Molecular Diversity · first author · DOI

2026 Anti-cancer and dual inhibitory potential of PI3K/AKT and Ras/MAPK/ERK signalling of a novel zinc (II) trinuclear complex with tetradentate schiff base ligand and azido ion in prostate adenocarcinoma: synthesis, in silico and in vitro evaluation
Abdul Wasai et al. · Journal of Molecular Structure · DOI

2025 Drug Repurposing for AML: Structure-Based Virtual Screening and Molecular Simulations of FDA-Approved Compounds with Polypharmacological Potential
Mena Abdelsayed et al. · Biomedicines · Link

2025 Computational screening of natural products as tryptophan 2,3-dioxygenase inhibitors: Insights from CNN-based QSAR, molecular docking, ADMET, and molecular dynamics simulations
Y. Boulaamane et al. · Computers in Biology and Medicine · first author · DOI

2025 Virtual Screening and Identification of Natural Molecules as Promising Quorum Sensing Inhibitors against Pseudomonas aeruginosa
Badr-Edine Sadoq et al. · Letters in Applied NanoBioScience · Link

2024 Computational Investigation of Phytochemicals from Aloysia citriodora as Drug Targets for Parkinson's Disease-Associated Proteins
Y. Boulaamane et al. · ChemistrySelect · first author · DOI

2024 Metal and Metal Oxide Nanoparticles: Computational Analysis of Their Interactions and Antibacterial Activities Against Pseudomonas aeruginosa
Badr-Edine Sadoq et al. · BioNanoScience · DOI

2024 Computational exploration of acefylline derivatives as MAO-B inhibitors for Parkinson's disease: insights from molecular docking, DFT, ADMET, and molecular dynamics approaches
Ali Irfan et al. · Frontiers in Chemistry · DOI

2024 In silico Discovery of Dual Ligands Targeting MAO-B and AA2AR from African Natural Products Using Pharmacophore Modelling, Molecular Docking, and Molecular Dynamics Simulations
Y. Boulaamane et al. · Chemistry Africa · first author · DOI

2024 Identification of Natural Inhibitors of SARS-CoV-2 Main Protease (Mpro) via Structure-Based Virtual Screening and Molecular Dynamics Simulations
Y. Boulaamane et al. · Letters in Applied NanoBioScience · first author · DOI

2024 Dendrobium nobile alkaloids modulate calcium dysregulation and neuroinflammation in Alzheimer's disease: A bioinformatic analysis
Iman Touati et al. · Pharmacological Research - Modern Chinese Medicine · DOI

2024 Antibiotic discovery with artificial intelligence for the treatment of Acinetobacter baumannii infections
Y. Boulaamane et al. · ASM mSystems · first author · DOI

2023 Identification of novel dual acting ligands targeting the adenosine A2A and serotonin 5-HT1A receptors
Iman Touati et al. · Journal of Biomolecular Structure and Dynamics · DOI

2023 Exploring natural products as multi-target-directed drugs for Parkinson's disease: an in-silico approach integrating QSAR, pharmacophore modeling, and molecular dynamics simulations
Y. Boulaamane et al. · Journal of Biomolecular Structure and Dynamics · first author · DOI

2023 Probing the molecular mechanisms of α-synuclein inhibitors unveils promising natural candidates through machine-learning QSAR, pharmacophore modeling, and molecular dynamics simulations
Y. Boulaamane et al. · Molecular Diversity · first author · DOI

2023 Chemical library design, QSAR modeling and molecular dynamics simulations of naturally occurring coumarins as dual inhibitors of MAO-B and AChE
Y. Boulaamane et al. · Journal of Biomolecular Structure and Dynamics · first author · DOI

2023 Insights into the Structure-Activity Relationship of Alkynyl-Coumarinyl Ethers as Selective MAO-B Inhibitors Using Molecular Docking
Y. Boulaamane et al. · Nova Science Publishers · book chapter, first author · Link

2022 β-amino carbonyl derivatives: Synthesis, Molecular Docking, ADMET, Molecular Dynamic and Herbicidal studies
Suneeta Bhandari et al. · ChemistrySelect · DOI

2022 In silico studies of natural product-like caffeine derivatives as potential MAO-B inhibitors/AA2AR antagonists for the treatment of Parkinson's disease
Y. Boulaamane et al. · Journal of Integrative Bioinformatics · first author · DOI

2022 Structural exploration of selected C6 and C7-substituted coumarin isomers as selective MAO-B inhibitors
Y. Boulaamane et al. · Journal of Biomolecular Structure and Dynamics · first author · DOI

Training

Jun 2025	**Certified Peer Reviewer Course** Elsevier Researcher Academy
Aug 2024	**AI for Drug Discovery: Deep and Reinforcement Learning** Neovarsity
Aug 2024	**ALX AiCE, AI Career Essentials** ALX Africa
Jan 2024	**Advanced Machine Learning for Drug Discovery** Neovarsity
Sep 2023	**Molecular Machine Learning Foundation** Neovarsity
Mar 2022	**Data Manipulation in Python: Python, NumPy and Pandas** Udemy
Feb 2021	**Computer-Aided Drug Design** Udemy

Projects

Learn CADD. Interactive guide to docking, pharmacophores, QSAR, molecular dynamics, and ADMET.
Assayer. Catalogue of 3,973 chemistry tools and 27 protocols, including UniProt checks and PDB choice.
BioLatent. Registry of embeddings for molecules, proteins, complexes, reactions, and nucleic acids.
SARScope. Experimental structure–activity reports from a ChEMBL target: series, activity cliffs, and QSAR checks.